

Reviewing Peer Reviewing

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Intended Learning Outcomes

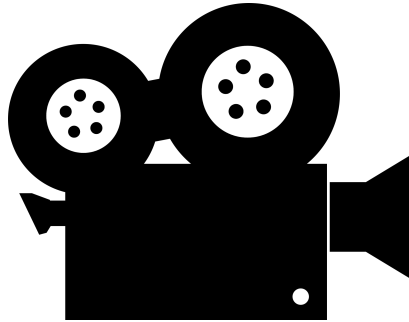
- Understand the history of peer reviewing
- Discuss the pros and cons of the current reviewing system

What are scientific articles for?

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- Communication of scientific knowledge
- But demonstrate productivity (allowing to get grants, jobs, promotions)

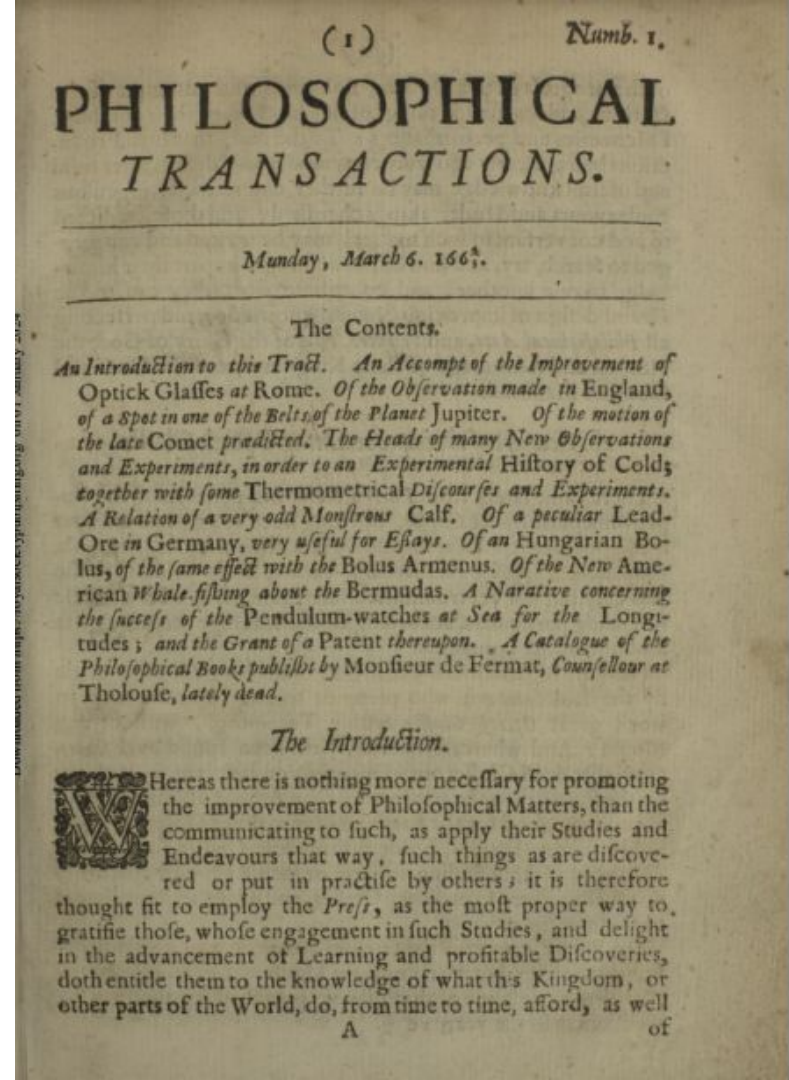
Peer review can be annoying



Peer Review

Supposedly, reviewers have been used when needed (external expert) since day 1, but a formal process appeared only 90 years later.

the aim was generally to protect the reputation of the scientific society that published the journal rather than to assess the merits of articles



How dare you peer review me!

Well into the twentieth century, many renowned scientists went their entire careers without having a paper refereed—and were not always enthusiastic when introduced to the practice. In 1936, for instance, Albert Einstein was extremely offended when he learned that the editor of *Physical Review* had sent his submitted paper to an external referee. In a terse note to the editor, John Tate, Einstein wrote that he and his co-author

had not authorized you to show [our manuscript] to specialists before it is printed. I see no reason to address the—in any case erroneous—comments of your anonymous expert. On the basis of this incident I prefer to publish the paper elsewhere.⁹

D. Kennefick, 'Controversies in the history of the radiation reaction problem in general relativity', in *The expanding worlds of general relativity* (H. Goenner et al.), pp. 207–234 (Springer, New York, 1999), at pp. 207–209

Pros and cons of peer reviews
(and are there different ways to do it?)

Peer review is new(ish)



CREDIBILITY, PEER REVIEW, AND *NATURE*, 1945–1990

by

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This paper examines the refereeing procedures at the scientific weekly *Nature* during and after World War II. In 1939 former editorial assistants L. J. F. Brimble and A. J. V. Gale assumed a joint editorship of *Nature*. The Brimble–Gale era is now most famous for the editors’ unsystematic approach to external refereeing. Although Brimble and Gale did sometimes consult external referees, papers submitted or recommended by scientists whom the pair trusted were often not sent out for further review. Their successor, John Maddox, would also print papers he admired without external refereeing. It was not until 1973 that editor David Davies made external peer review a requirement for publication in *Nature*. *Nature*’s example shows that as late as the 1960s a journal could be considered scientifically respectable even if its editors were known to eschew systematic external peer review.

equipment, and to Dr. G. E. R. Deacon and the captain and officers of R.R.S. *Discovery II* for their part in making the observations.

¹ Young, F. B., Gerrard, H., and Jevons, W., *Phil. Mag.*, **40**, 149 (1920).

² Longuet-Higgins, M. S., *Mon. Not. Roy. Astro. Soc., Geophys. Supp.*, **5**, 285 (1949).

³ Von Arx, W. S., Woods Hole Papers in Phys. Oceanog. Meteor., **11** (3) (1950).

⁴ Ekman, V. W., *Arkiv. Mat. Astron. Fysik. (Stockholm)*, **2** (11) (1905).

MOLECULAR STRUCTURE OF NUCLEIC ACIDS

A Structure for Deoxyribose Nucleic Acid

WE wish to suggest a structure for the salt of deoxyribose nucleic acid (D.N.A.). This structure has novel features which are of biological interest.

A structure for nucleic acid has already been proposed by Pauling and Corey¹. They kindly made their manuscript available to us in advance of publication. Their model consists of two intertwined chains, with the phosphates near the axis, and the bases on the outside. In our model this structure is unsatisfactory for two reasons. (1) We believe that the material which gives the X-ray diagrams is the salt, not the free acid

is a residue on each chain every 3.4 Å. in the z-direction. We have assumed an angle of 36° between adjacent residues in the same chain, so that the structure repeats after 10 residues on each chain, that is, after 34 Å. The distance of a phosphorus atom from the fibre axis is 10 Å. As the phosphates are on the outside, cations have easy access to them.

The structure is an open one, and its water content is rather high. At lower water contents we would expect the bases to tilt so that the structure could become more compact.

The novel feature of the structure is the manner in which the two chains are held together by the purine and pyrimidine bases. The planes of the bases are perpendicular to the fibre axis. They are joined together in pairs, a single base from one chain being hydrogen-bonded to a single base from the other chain, so that the two lie side by side with identical z-co-ordinates. One of the pair must be a purine and

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J. D. WATSON
F. H. C. CRICK

Medical Research Council Unit for the
Study of the Molecular Structure of
Biological Systems,
Cavendish Laboratory, Cambridge.
April 2.

¹ Pauling, L., and Corey, R. B., *Nature*, **171**, 346 (1953); *Proc. U.S. Nat. Acad. Sci.*, **39**, 84 (1953).

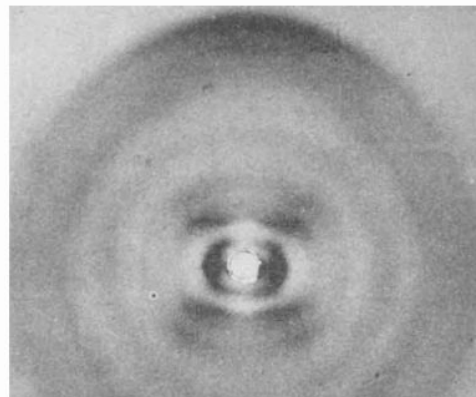
² Furberg, S., *Acta Chem. Scand.*, **6**, 634 (1952).

³ Chargaff, E., for references see Zamenhof, S., Brawerman, G., and Chargaff, E., *Biochim. et Biophys. Acta*, **9**, 402 (1952).

⁴ Wyatt, G. R., *J. Gen. Physiol.*, **36**, 201 (1952).

⁵ Astbury, W. T., *Symp. Soc. Exp. Biol.*, **1**, Nucleic Acid, 66 (Camb. Univ. Press, 1947).

⁶ Watson, J. D., and Crick, F. H. C., *Biochim. et Biophys. Acta*, **9**, 402 (1952).



Peer review is dead(ish)

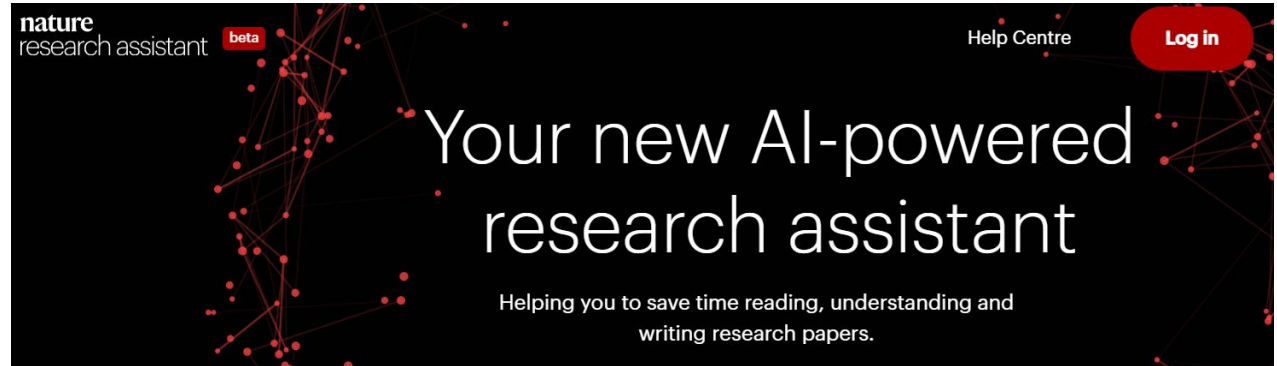
CAREER COLUMN | 04 March 2025

Three AI-powered steps to faster, smarter peer review

Tired of spending countless hours on peer reviews? An AI-assisted workflow could help.

By [Dritjon Gruda](#) 

Who is going to review in 10 years?

A dark-themed banner for the Nature Research Assistant. On the left, there is a network diagram with red nodes and lines. The text 'nature research assistant' is in the top left, with a red 'beta' badge next to it. In the top right, there is a 'Help Centre' link and a red 'Log in' button. The main headline reads 'Your new AI-powered research assistant' in large white font. Below it, a subtitle says 'Helping you to save time reading, understanding and writing research papers.'

nature
research assistant **beta**

Help Centre **Log in**

Your new AI-powered research assistant

Helping you to save time reading, understanding and writing research papers.

Peer review is dead(ish)

That Crick and Watson paper gets rejected by AI for unsubstantiated claims btw ..



Overstated Claims

It has not escaped our notice that the specific pairing we have postulated immediately suggests a possible copying mechanism for the genetic material.

① *This statement goes beyond the data and model presented, as it infers a functional mechanism (copying mechanism for genetic material) directly from the proposed structural model, without experimental evidence demonstrating such a mechanism.*

<https://bsky.app/profile/tomerullman.bsky.social/post/3m4q6c5alc227>

You review all the time, it's called Journal club

Post-online? <https://pubpeer.com/>

Peer reviewer Oath: <https://f1000research.com/articles/3-271/v2>

Pros and Cons

- **Anonymous reviews**
- **Signed Reviews**
- **Publish reviews and responses**
- **Publish then review --**

<https://elifesciences.org/about/peer-review>

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